

Runbook: Failed Load Recovery - Harbor Data Lake to Sentinel TM

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1. Pipeline Overview

Source files from upstream systems land in the S3 landing bucket (nwb-hdl-landing). A Step Functions state machine (hdl-daily-ingest) runs an AWS Glue job per feed that validates the file, writes it to the staging zone, then merges it into the curated zone (Apache Iceberg tables). At 06:30 ET the Sentinel extract job reads the curated zone, and Sentinel TM batch scoring starts at 07:00 ET.

2. When to Use This Runbook

Use this runbook whenever a feed's Glue job fails, a Step Functions execution is aborted, or a load completes with a validation error. For files that arrive late, start with NWB-RB-402.

3. Recovery Steps

All steps are mandatory and must be performed in order. Record each step in the incident ticket.

- 1 **Acknowledge** the PagerDuty alert and open an incident ticket in #hdl-incidents.
- 2 **Identify the failed stage** from the hdl-daily-ingest Step Functions execution history and Glue job logs.
- 3 **Place a hold** on the Sentinel extract for the affected feed by setting `hold_flag = true` in the `hdl_control.feed_status` table.
- 4 **Truncate the affected business-date partition in staging** before any rerun. Never rerun into a partially written partition - doing so creates duplicate or partial records.
- 5 **Rerun the Glue job** for the affected feed and business date.
- 6 **Reconcile row counts and hash totals** against the source file's trailer record. Counts and amount hash totals must match exactly for every feed (wire, ACH, cash, card, customer).
- 7 **Run the duplicate-key check** on `transaction_id + business_date` in the curated table. The expected result is zero duplicates.
- 8 **Release the hold** (`hold_flag = false`), trigger the Sentinel extract, and confirm the record count ingested by Sentinel TM matches the curated count.
- 9 **If the same-day scoring window was missed**, notify the FCC Operations Manager so a lookback run can be requested per NWB-POL-101 section 6.
- 10 **Update and close** the incident ticket; schedule a postmortem for any Sev-1 or Sev-2.

4. Common Failure Causes

Symptom	Likely cause	Note
Glue job fails with OutOfMemory	Feed volume spike (month-end, holidays)	Rerun with G.2X workers
S3 SlowDown / 503 errors	Request throttling on landing prefix	Glue auto-retry may leave a partial partition - step 4 is critical
Trailer mismatch	File truncated in transit	Request re-send from upstream owner